

Activity 1: Model Providers and Recent Models

Provider	Model	Release	Knowledge Cutoff	Cost (Input/Output per 1M tokens)	Context	Type	When to use
OpenAI	GPT-5.5	Jun 2026	Dec 2025	\$5/\$30	1M	General	Reasoning, coding, agents
Anthropic	Claude Opus 4.8	Jun 2026	Jan 2026	\$5/\$25	1M	Reasoning	Long docs
Google	Gemini 3.1 Pro	May 2026	Jan 2025	\$2/\$12	1M	Multimodal	Enterprise AI
Meta	Llama 4 Scout	Apr 2025	Aug 2024	Open weights	10M	Open	Self-hosting
DeepSeek	DeepSeek V4	Feb 2026	Late 2025	~\$0.8/\$2.4	1M	Reasoning	Low-cost coding
Alibaba	Qwen 3.5 Plus	May 2026	2026	~\$0.6/\$2.4	1M	General	Multilingual
MiniMax	MiniMax M3	Feb 2026	Jan 2026	~\$1/\$5	1M	Long context	Large docs
Moonshot AI	Kimi K2.5	Feb 2026	Apr 2025	~\$1/\$5	1M	Reasoning	Research
Zhipu AI	GLM-5.2	Jun 2026	2025	\$1.4/\$4.4	1M	General	Chinese AI
xAI	Grok 4.1	Late 2025	Nov 2024	Varies	2M	General	Realtime analysis

Activity 2: Understanding Transformers

(a) Self-Attention Mechanism

Self-attention allows every token in a sequence to determine how much attention to pay to every other token. Each token is transformed into Query (Q), Key (K), and Value (V) vectors. Attention scores are computed using the similarity between Q and K, normalized with a softmax function, and applied to the Value vectors. Multi-head attention performs this process in parallel to capture different relationships in the input.

(b) Vector Embeddings

Vector embeddings convert words or tokens into dense numerical vectors. Tokens with similar meanings are represented by similar vectors, enabling the model to understand semantic relationships rather than treating words as simple identifiers.

(c) Feed Forward Neural Network (FFN)

Following the attention layer, each token passes independently through a feed forward neural network, typically consisting of two linear layers with a GELU or ReLU activation in between. The FFN extracts higher-level features and improves the model's ability to learn complex patterns.

Transformer Flow:

Input Tokens → Embeddings → Positional Encoding → Multi-Head Self-Attention → Add & Normalize → Feed Forward Network → Add & Normalize → Output.